

About the project

This project was developed by Emilio Barragán, a Computer Science student at Tecnológico de Monterrey, as part of a personal, non-school-related project in March 2025.

The main idea of this project was to extract and analyze data directly from a website about cars or vehicles in general. My main idea was to find a dataset that would allow me to determine the most popular car brand in the world. However, I discovered a website called Fuel Economy, which is the official source of fuel economy information in the US.

Data Extraction

Once I found the source where I was going to extract data, I began looking for the endpoints and the data I could use to make some interesting EDA and data Visualization. I found two XML-based endpoints, each containing different but related vehicle information:

- **Vehicle table:** included general information about all registered vehicles.
- **ympgVehicle table:** contained user-reported average miles per gallon (mpg) and the state of origin.

Both endpoints used a single parameter (vehicleid), which made it possible to link records between tables.

EDA

Once the data was successfully extracted, it was stored in 2 separate CSV files and then used for a simple EDA. The dataset included 48,800 vehicles and over 51,000 shared mpg records.

As I mentioned, my main idea was to determine what the most popular brand in the world is, but since the dataset only contains US data, I could find out what the most popular car brands were in every state, and the results were the following:

CA	Toyota	1034
TX	Toyota	454
FL	Toyota	401
MI	Ford	392
NY	Toyota	306
IL	Ford	244

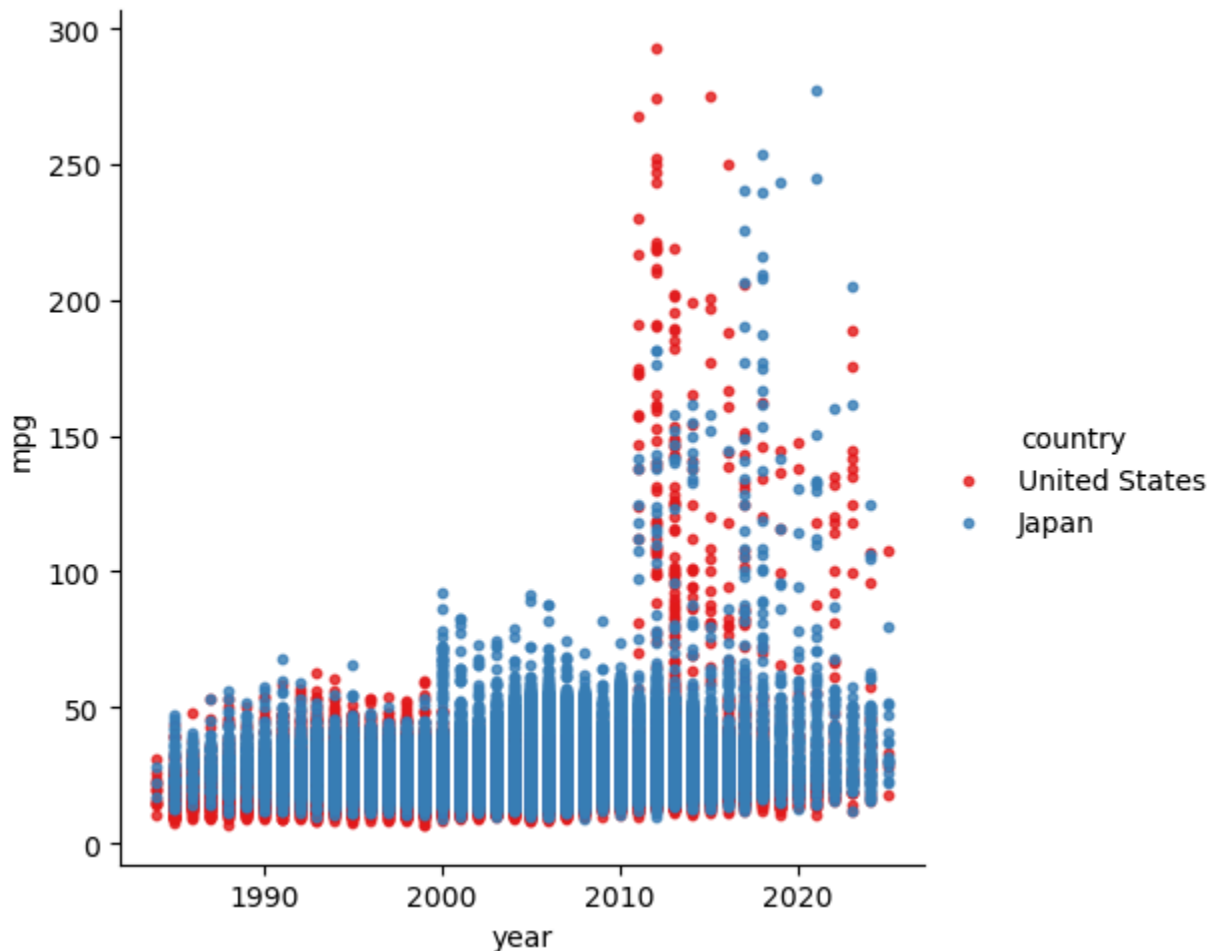
And since this data also included the model of the car, by changing the query (in Pandas), I also discovered the most popular models in every state:

CA	Prius	208
TX	Civic	91
FL	Civic	78
PA	Civic	70
IL	Civic	68

NY Civic 65

It's interesting how the most popular brand is Toyota, but the most popular model is the Civic from Honda.

But this EDA wasn't just about simple queries in Python syntax; I also wanted to make a comparison between American and Japanese vehicles, which are the most popular in the US. In this case, I wanted to use the mpg, which is something the users register in the Vehicle Fuel web page



Coding and packages

The coding language used in this project was mainly Python in a Jupyter Notebook environment. And like most data science projects, I used Pandas for data manipulation and to create queries (like SQL queries), as well as Seaborn to create graphs. For the data extraction part, I used BeautifulSoup and requests to access the XML code and data in the endpoints.

Conclusion

This project demonstrated how real-world data can be extracted and analyzed directly from web sources using Python. Although the dataset was limited to U.S. vehicles, the analysis provided

insights into brand and model popularity, as well as comparative performance between American and Japanese manufacturers. The project also strengthened my understanding of XML parsing, data cleaning, and exploratory data analysis in Python.

Data source: <https://www.fueleconomy.gov/feg/ws/index.shtml>

Github: https://github.com/Emiliobg06/Cars_inv